

Superpave for Bitumen Binder Characterisation

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For some years now, NZTA has updated bitumen testing and specifications.

§ “For many years New Zealand bitumen was based on an empirical Penetration based specification. However, changes to the crude oil feedstock used to manufacture bitumen, plus the importation of bitumens into New Zealand from other refiners who use other crude oil feedstock has meant that the empirical Penetration test could no longer be used to infer performance. This has led to the adoption of a performance-based specification for asphalt binders.”

§ It's all about SuperPave!

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Superpave

§ Superior Performing Asphalt Pavements

§ Performance-based system for

- specifying materials
- performing mix design

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Overview of Old Specifications

§ Bitumen Characterisation Testing

- Penetration Test at 25 °C
- Viscosity Test at 60 °C and 135 °C
- Ductility Test at 25 °C
- Softening Point

Ageing is only simulated for short term by Rolling Thin Film Oven Test (RTFOT). No consideration for Long Term ageing in the Old Specifications.

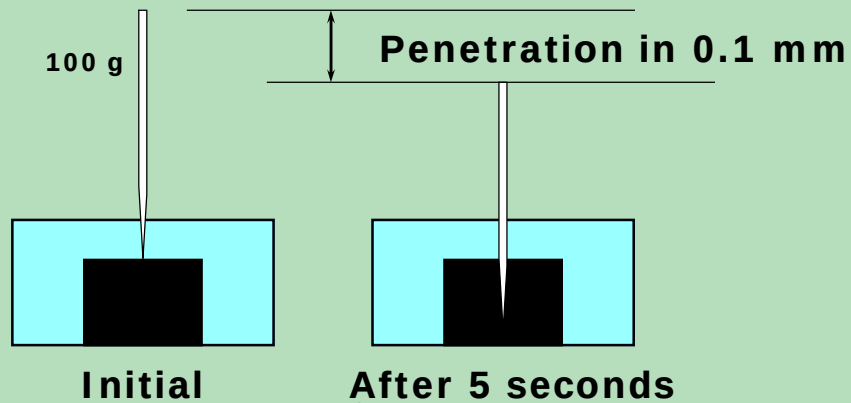
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Penetration Testing

§ Sewing machine needle

§ Specified load, time, temperature



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Penetration Specification

§ Five Grades

- 40 - 50
- 60 - 70
- 80 - 100
- 120 - 150
- 200 – 300 (USA)

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Viscosity Graded Specifications

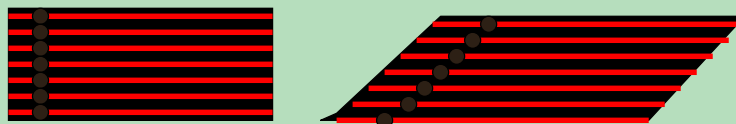
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Viscosity: the ratio between the applied shear stress and the rate of shear strain.

$$\eta = \tau / (d\gamma/dt)$$

The higher the viscosity, the harder the material.



Units: stress*time or Pa x sec = 10 Poise (1 Poise = 0.1 Pa x sec)

Dividing by density: Pa x sec / (kg/m³) = m²/sec (1 Stoke = 1 cm²/sec)

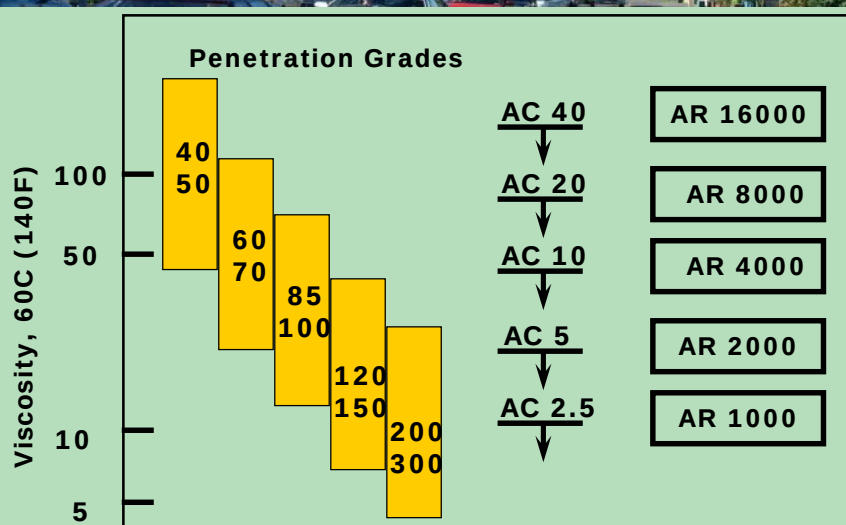
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Viscosity Grading System

Bitumen Type	AC 2.5	AC 40
Vis. (Poises), 60°C	250 ± 50	4000 ± 800
Visc. (Poises), 135 °C	80 +	300 +
Penetration	200 +	20 +
Ductility	100 +	10 +

Temperature of the test is constant regardless of the actual pavement temperature. It does not cover the entire temperature range.

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- § Penetration Grade is the oldest grading system. It has several disadvantages. NZ and some few countries have still been using it.
- § Viscosity Grade is step better than the penetration grade and is available for original unaged binder and for short term aged binder. This system is still having some problems as it will be discussed later. Australia is currently using this system.
- § Performance Grade is the product of the Superpave and it will be discussed in full details in later sections.

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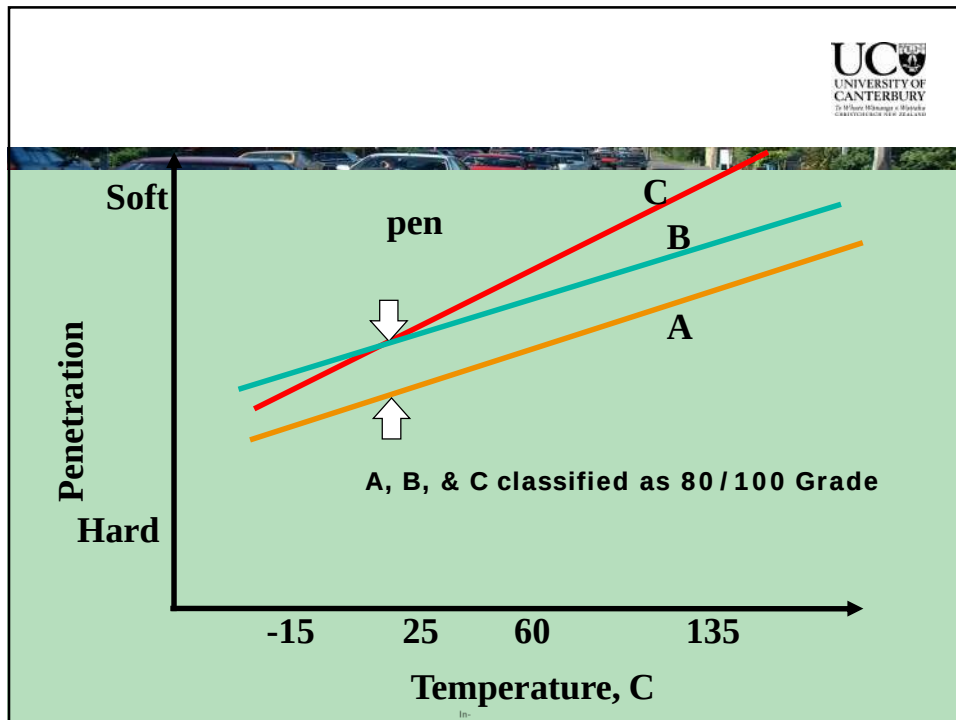
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Drawback of the Old Specification System

- § Penetration, Softening point and Ductility are **empirical** tests.
- § They are not directly correlated with field performance
- § With empirical tests, pavement performance experience is required before the test results can yield any meaningful information.
- § Give information at specific pavement temperatures and do not give information for the entire range of pavement temperatures.

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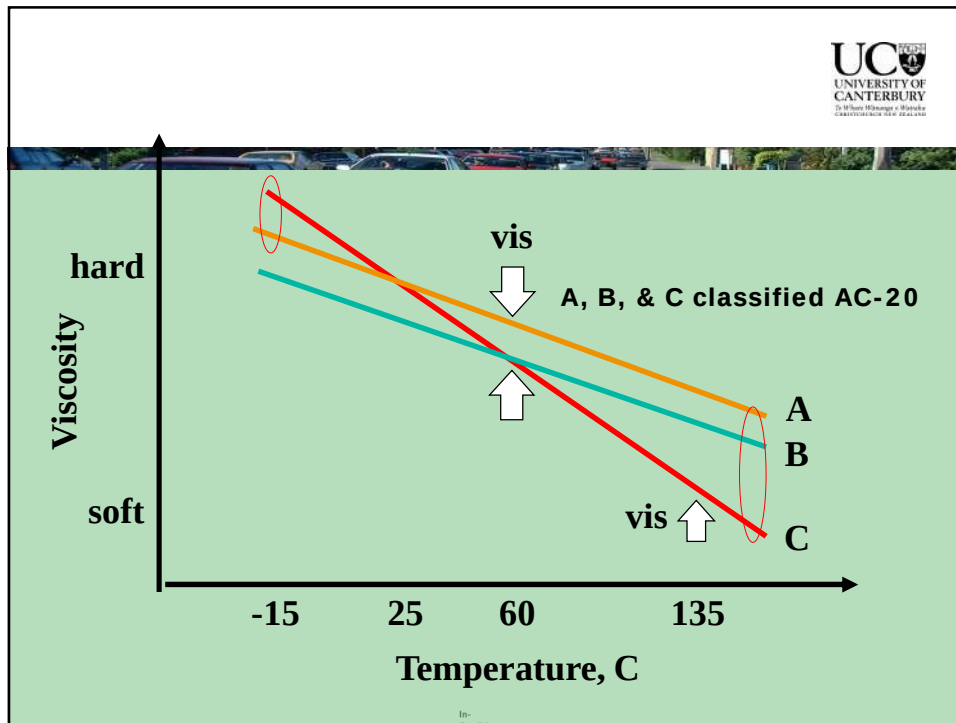


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- § Bitumens A, B, and C are three different sources of bitumen.
- § According to the penetration classification system they are all classified as 80/100 grade.
- § However, the three binders behave totally different with binder C showing very high sensitivity to temperature change compared to binders A and B.

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To Whom It May Concern

- § The same Problem was found in the viscosity grade system. A, B, and C shows the same viscosity grade
- § While bitumens A and B display the same temperature dependency, they have much different consistency at all temperatures.
- § Bitumens A and C have the same viscosity at low temperatures, but remarkably different viscosity at high temperature. Bitumen B has the same viscosity at 60C, but shares no other similarities with bitumen C.

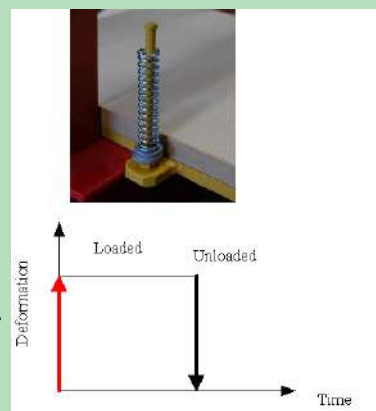
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Because these bitumens meet the same grade specifications, one might erroneously expect the same characteristics during construction and the same performance during hot and cold weather conditions.

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Viscoelastic Behaviour of Bitumen

- § Elastic Behaviour
- § Material will deform instantaneously upon loading and will rebound back immediately upon unloading.
- § Elastic Behaviour is time independent
- § This resembles the behaviour of asphalt mix at very low temperature



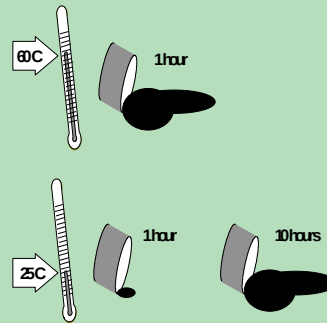
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Time-Temperature Dependency

§ Bitumen and asphalt materials are both time and temperature dependent.

§ For example, the effect at 60°C and time duration of 1 hr is similar to the effect at 25°C but with time duration 10 Hours.

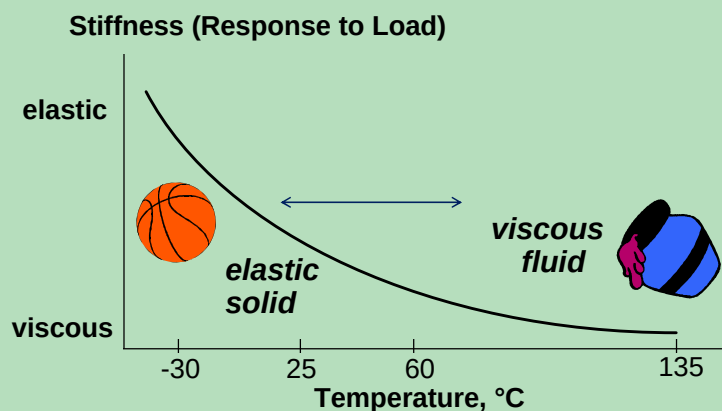
§ How can we benefit from that phenomenon? (See BBR Test?)



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Effect of Temperature on Bitumen / Asphalt



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High Temperature Behavior

§ High in-service temperature

- Desert climates
- Summer temperatures



Viscous Liquid

§ Sustained loads

- Slow moving trucks
- Intersections

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Pavement Behavior (Warm Temperatures)

§ Permanent deformation (rutting)

§ Mixture is plastic

§ Depends on bitumen source, additives, and aggregate properties

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Permanent Deformation



Effect of warm weather and traffic

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Distortion - Rutting

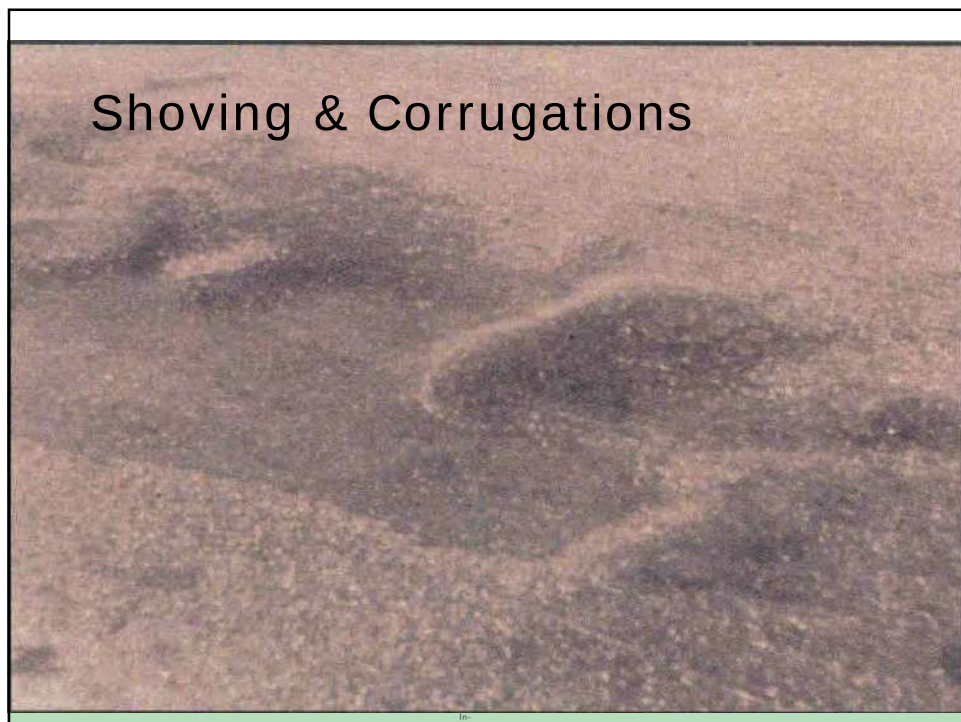


- **Vertical Displacements**
- **Lateral Displacements**

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Rutting Mechanisms

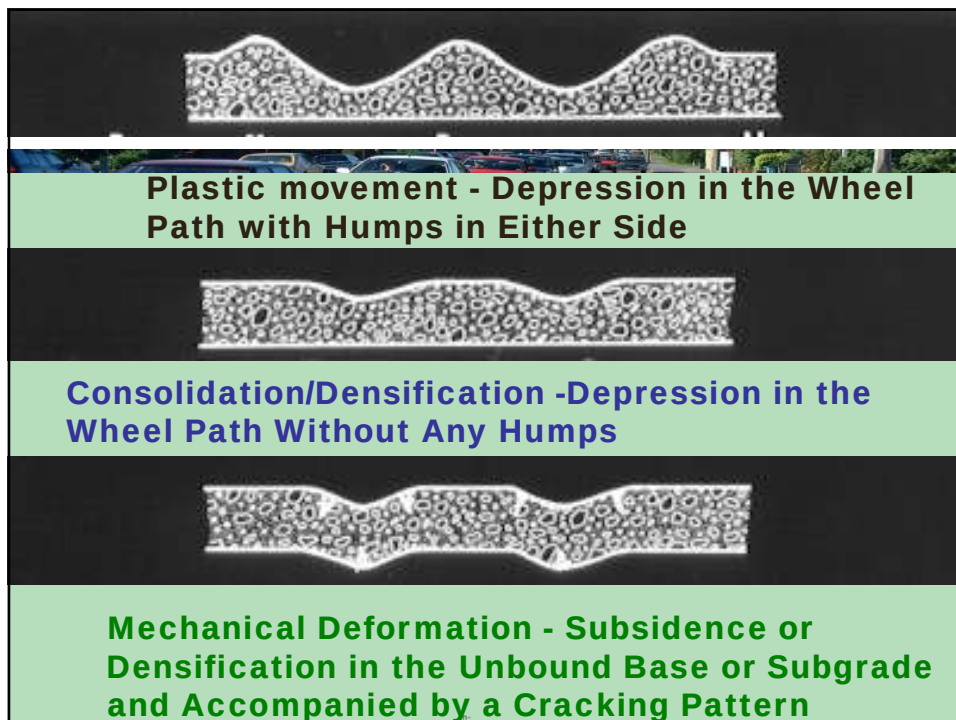
§ Potential Primary Causes:

- Plastic movement of HMA in hot weather
- Inadequate compaction of HMA during construction

§ Potential Secondary Causes:

- Plastic movement in other layers
- Inadequate compaction in other layers

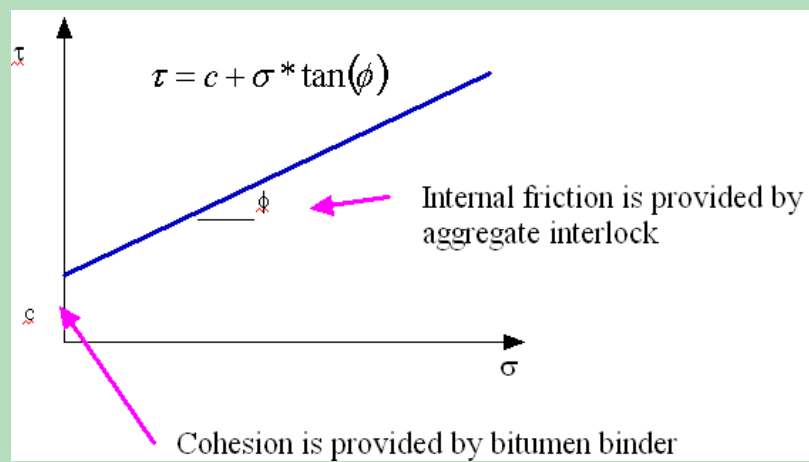
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- § Both aggregate and binder play a role in permanent deformation behaviour.
- § Binder must be stiff and elastic to provide high stiffness and cohesion for the aggregate particles.
- § Aggregates must have high internal friction provided by angular rough surfaces to provide the necessary stability.

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Low Temperature Behaviour

§ Low Temperature

- Cold climates
- Winter

§ Rapid Loads

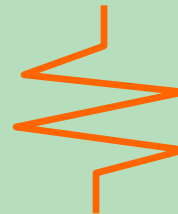
- Fast moving trucks



Elastic Solid

Hooke's Law

$$\sigma = E \epsilon$$
$$\bullet = \bullet \bullet$$



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Pavement Behavior (Low Temperatures)

§ Thermal cracks

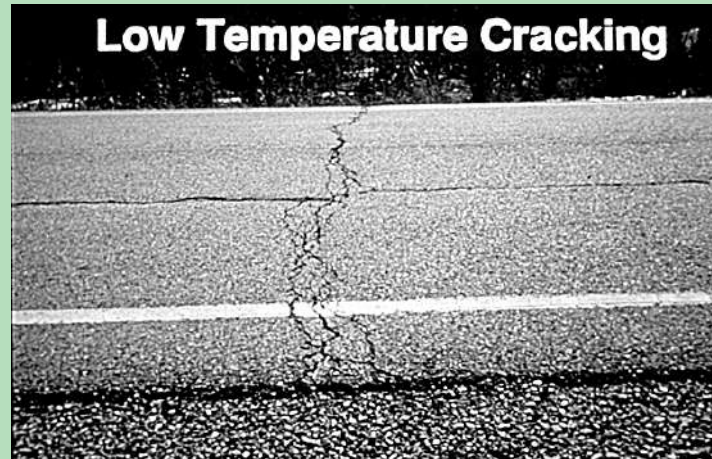
- Stress generated by contraction due to drop in temperature
- Crack forms when thermal stresses exceed ability of material to relieve stress through deformation
- Material is brittle

§ Depends on source of bitumen and aggregate properties

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Thermal Cracking



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§ Usually thermal cracks occur at a later stage of pavement life when bitumen is likely to be hardened by oxidation. Thus in the Superpave testing, the bitumen must subject to both short term and long term ageing.

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Low Temperature Behaviour

§ Low Temperature

- Cold climates
- Winter

§ Rapid Loads

- Fast moving trucks

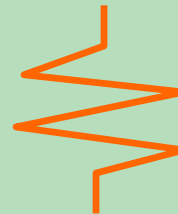


Elastic Solid

Hooke's Law

$$\sigma = E \epsilon$$

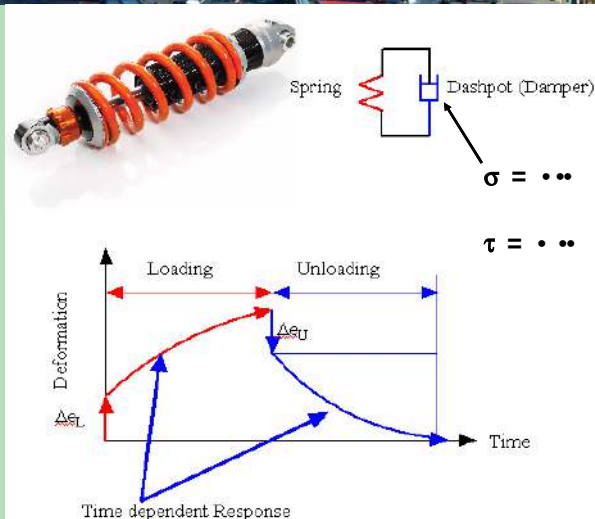
$$\dot{\sigma} = \dots$$



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Intermediate Temperature Behaviour



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Intermediate Temperature Behaviour

At intermediate temperature, bitumen displays combinations of both elastic and viscous response similar to spring and dashpot model shown.

In hot mix asphalt, the spring represents the immediate elastic response of both the bitumen and the aggregate. The dashpot symbolizes the slower/delayed, viscous reaction of the bitumen, particularly in warmer temperatures.

Most of the response is elastic or viscoelastic, (recoverable with time), while some of the response is plastic and non-recoverable.

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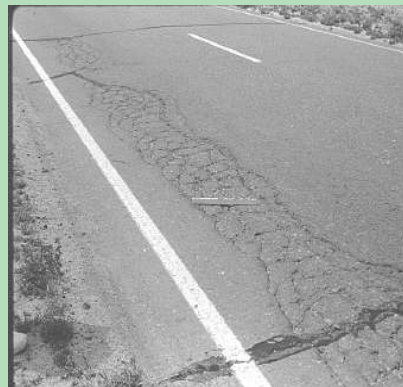
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Fatigue Cracking

§ Usually fatigue cracks are associated with intermediate temperatures.

§ Fatigue cracks are caused by repeated tensile strains at the bottom of the asphalt mix.

§ Mix needs to be Soft and Elastic to resist fatigue cracking



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Ageing Behaviour

- § Bitumen is an organic material that subject to oxidation under the effect of oxygen and heat.
- § Oxidation will cause bitumen to become brittle
- § Short Term Ageing: A considerable amount of oxidative hardening occurs at the hot mix facility because of the elevated temperatures and thin bitumen coatings (Simulated in the lab by RTFOT).
- § Long Term Ageing: This is the continuous oxidation and hardening that occurs to bitumen over the pavement service life (Simulated in the lab by PAV).

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Ravelling due to excessive hardening



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New Superpave Binder Specifications



- § The Superpave binder specification is intended to improve performance by
- § reducing the potential for the bitumen to contribute to permanent deformation, low temperature cracking, and early fatigue cracking in HMA pavements

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Why New Specifications?



- § Increased traffic and loadings
- § Supply sources
- § Use of Reclaimed Asphalt Pavements (Recycling) (RAP)
- § Personnel experience

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Superpave Binder Specifications



- § Permanent deformation
- § Excessive ageing from volatilization
- § Fatigue cracking
- § Low temperature cracking
- § Pumping and handling

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Benefits of Superpave

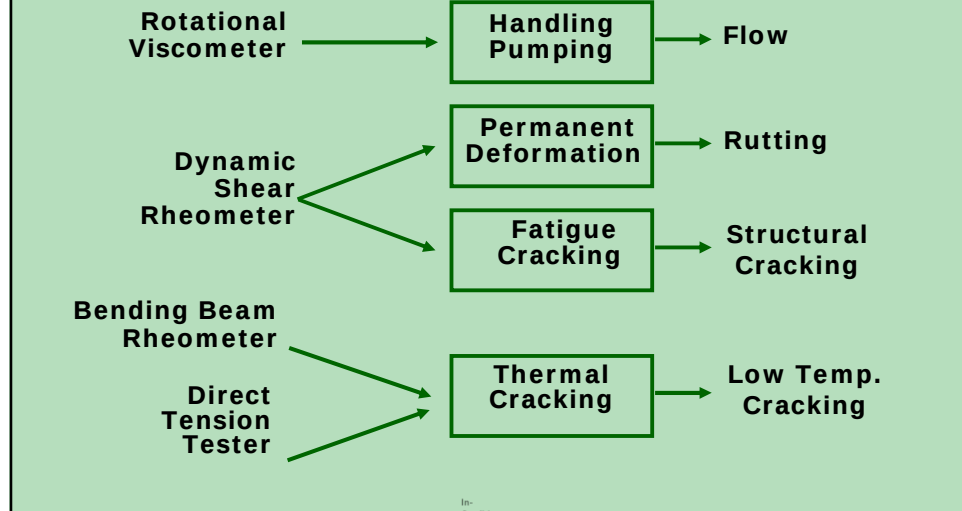


- § Better design and performance
- § Little measurable rutting
- § Minor changes to construction practices
- § Longer pavement life cycle

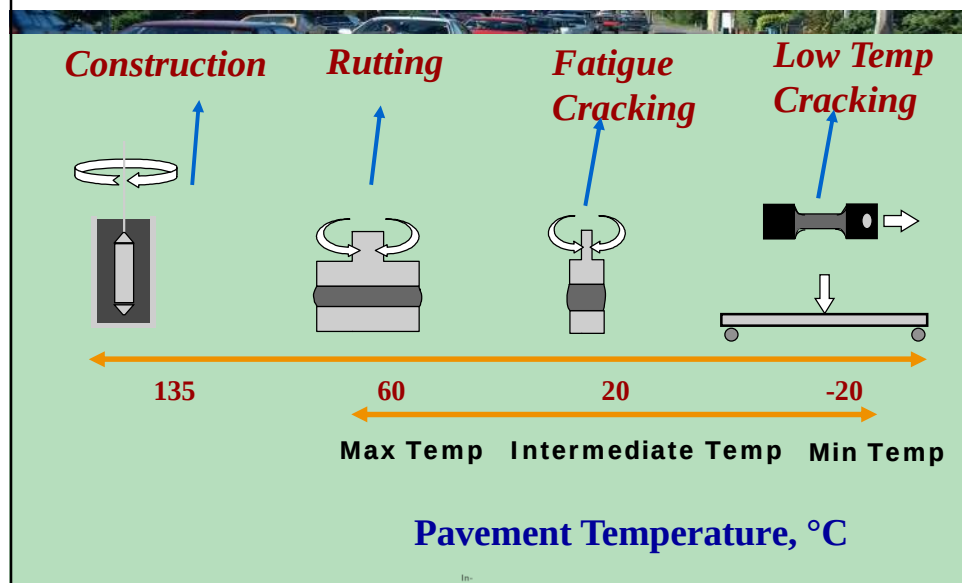
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Test Equipment Performance Property



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How Superpave Differs?

- § Lab compactor
- § Specimen size
- § Gradation tends to be coarser
- § Often requires aggregates with more fractured faces
- § **New bitumen binder specification**
- § Others

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How Superpave is Similar

- § Same production and placement problems
- § Difficulties in meeting compaction requirements
- § Others

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Short and Long Term Ageing Simulations

§ Aged binder

- Since long term performance problem, include:

Short term ageing

Long term ageing

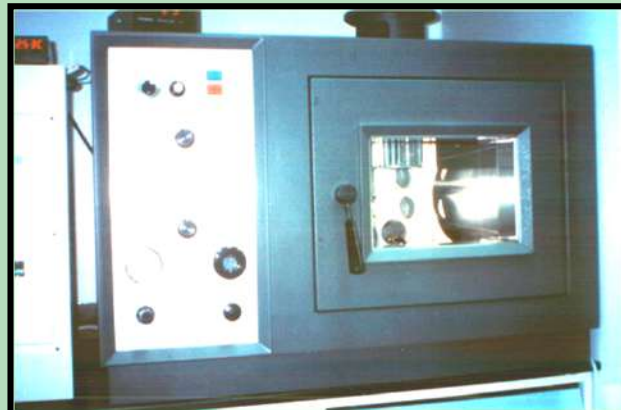
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Short Term Binder Ageing

§ Rolling Thin Film Oven Test (RTFOT)

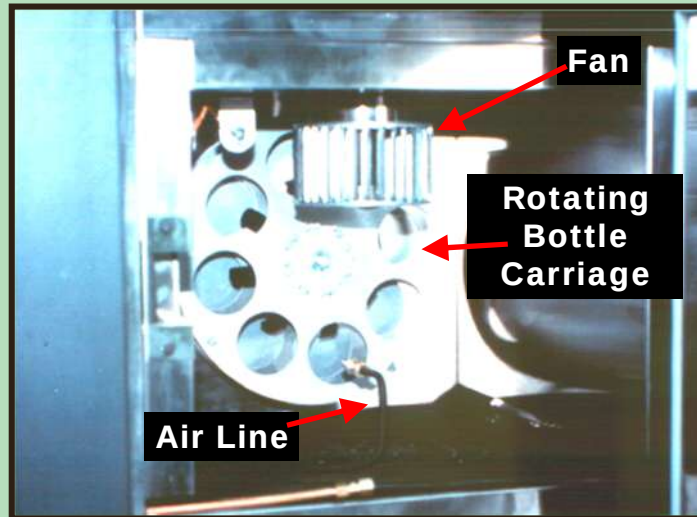
- Simulates ageing from hot mixing and construction



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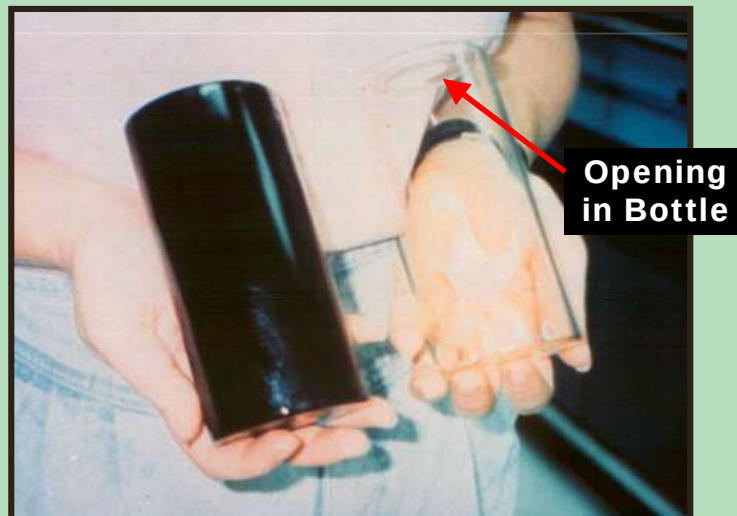
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Inside of RTFO



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Bottles Before and After Testing



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Testing

§ Calculate mass loss after RTFO

$$\text{Mass loss, \%} = \frac{\text{Original mass} - \text{Aged mass}}{\text{Original mass}} \times 100$$

Short term aged binder can be used for rutting characterisation tests (Majority of the rutting usually happens early on pavement life).

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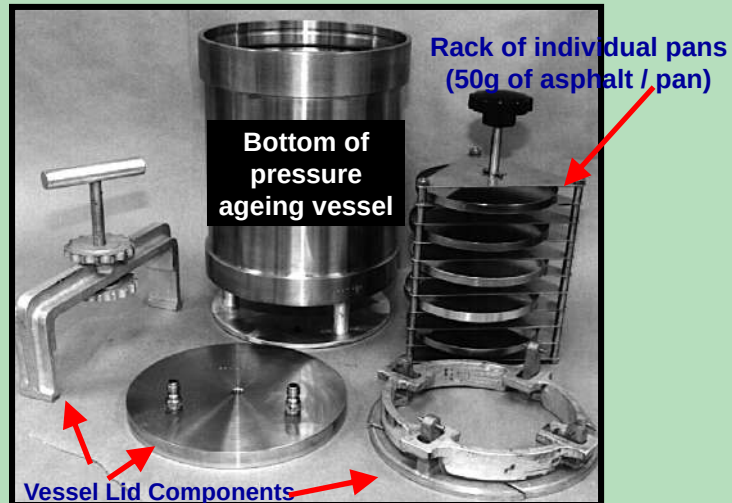
Pressure Ageing Vessel (Long Term Ageing)

- § Simulates ageing of bitumen binder for 7 to 10 years
- § 50 gram of the short term aged sample is aged for 20 hours in the PAV
- § Pressure of 2,070 kPa (300 psi)
- § At 90, 100 or 110 C

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Pressure Ageing Vessel



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Pressure Ageing Vessel



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ROTATIONAL VISCOMETER



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ROTATIONAL VISCOMETER



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Viscosity Units

- § Absolute viscosity can be measured by Poise or Pa.s
- § 1 Poise = 100 cP
- § 1 Poise = 1 gram/cm²/s = 1 dyne.s/cm²
- § 1 Poise = 10⁻⁵ N.s/(10⁻⁴ m²) = 0.1 N.s/m² = 0.1 Pa.s
- § To convert, this equation is used:
- § 1000 cP = 1 Pa.s
- § 1 cP = 1 mPa.s
- § **We want viscosity @135°C < 3 Pa.sec**

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- § To resist rutting, a binder needs to be Stiff and Elastic.
- § To resist fatigue cracking, the binder needs to be Soft and Elastic.
- § The balance between these two requirements is a critical one.

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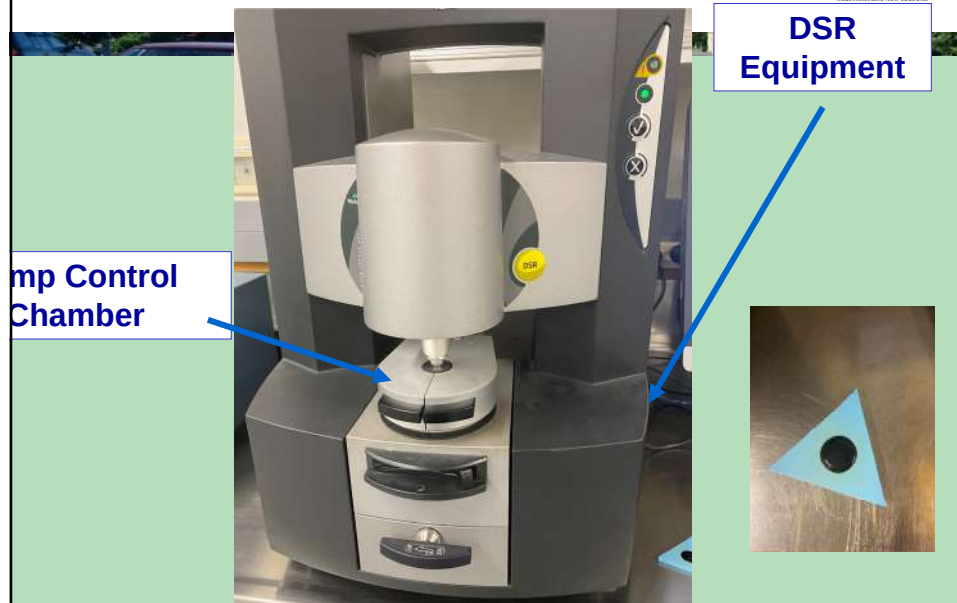
Dynamic Shear Rheometer (DSR)

- § At normal temperature, bitumen behaves as a viscoelastic material.
- § The Dynamic Shear Rheometer (DSR) is used to characterise the viscous and elastic behaviour of bitumen binders. It does this by measuring the viscous and elastic properties of a thin bitumen binder sample sandwiched between an oscillating and a fixed plate.
- § The DSR is used to characterise permanent deformation and fatigue properties of bitumen binders.

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Dynamic Shear Rheometer (DSR)



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**Parallel Plates
with Sample**

sample

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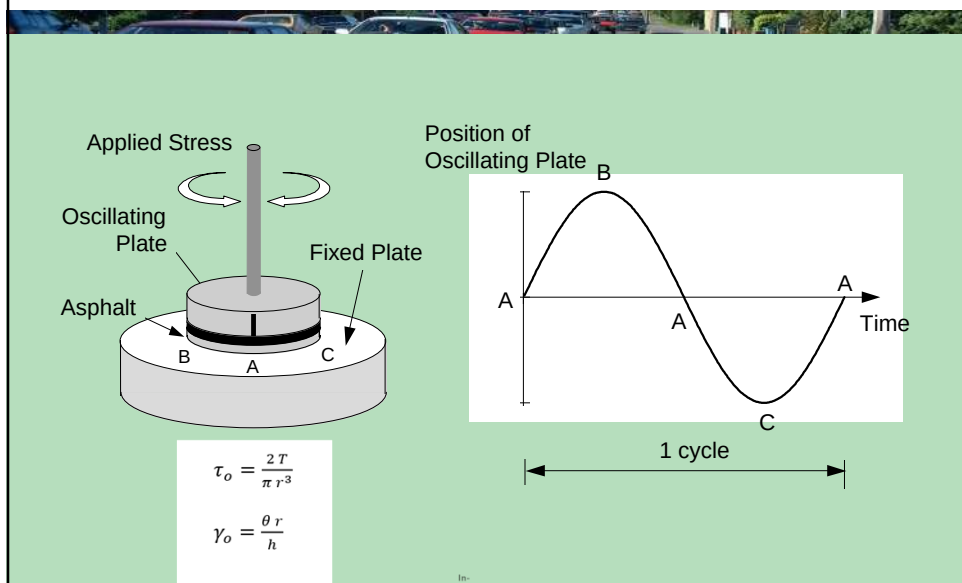


25 mm Plate with Sample

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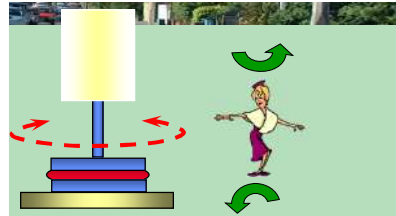
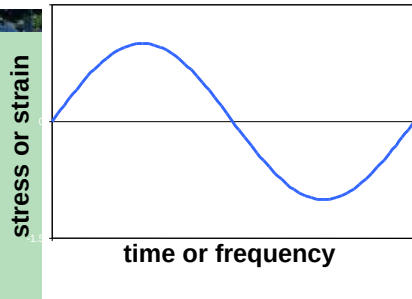


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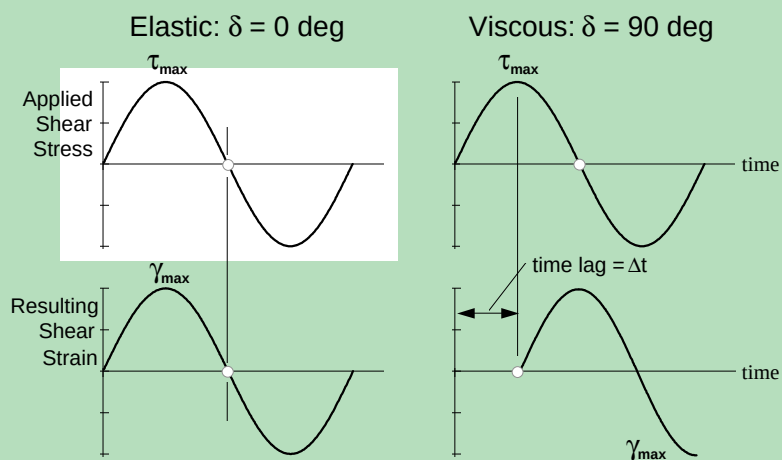
Dynamic Shear Rheometer



- Deformation or Stress applied sinusoidally
- Amplitude of Stress or Deformation & Frequency defined as per specification

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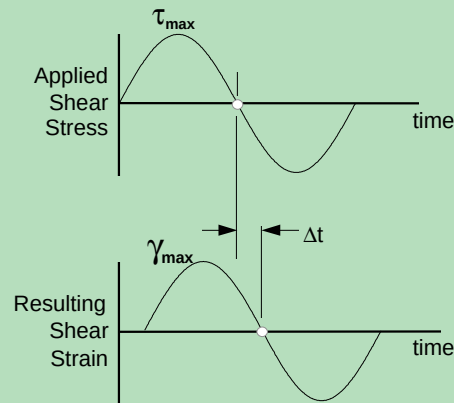
Elastic Versus Viscous Behaviour



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Viscoelastic Behaviour

Viscoelastic: $0 < \delta < 90^\circ$



$$G^* = \frac{\tau_{\max}}{\gamma_{\max}}$$

$\Delta t = \text{time lag} \Rightarrow \delta$

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- § Complex Shear Modulus (G^*) is the ratio of maximum shear stress (max) to maximum shear strain (max).
- § The phase angle (δ) is the time lag between the applied stress and the resulting strain.
- § For a perfectly elastic material, $\delta = 0$ (Low temperature)
- § For a perfectly Viscous material, $\delta = 90$ (High Temperature)
- § For Viscoelastic material $0 < \delta < 90$ (at service Temperature)

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Describing this viscoelastic behaviour

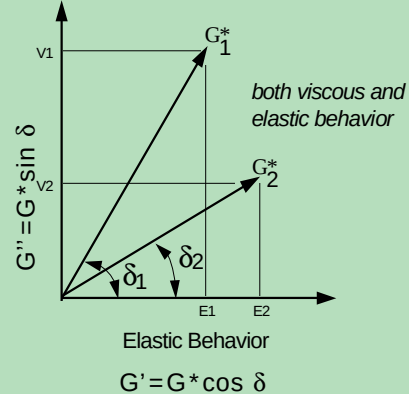
G^* is a measure of the total resistance of a material to deforming when repeatedly sheared.

Horizontal axis represent the elastic (recoverable deformation).

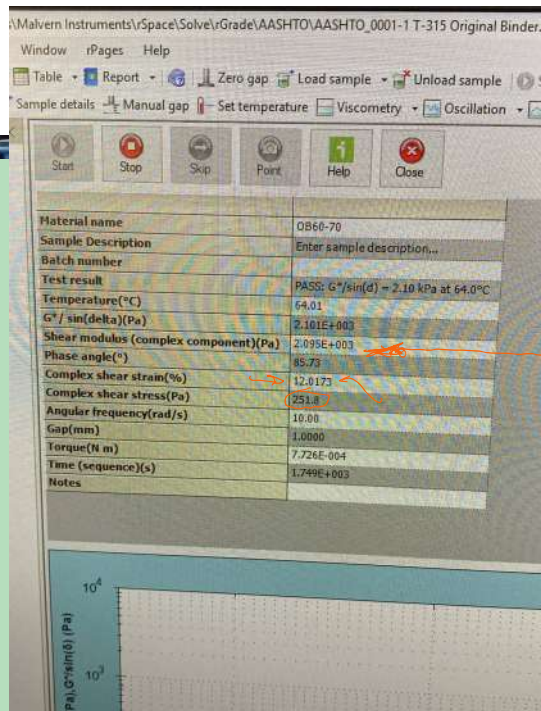
Vertical axis represents the viscous (non-recoverable)

Phase angle is the angle with the horizontal indicates the relative amounts of permanent and recoverable deformation.

Viscous Behavior



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High Temperature (Rutting) Parameter

§ The permanent deformation is caused by the dissipated energy W_c (work done) with every cycle of traffic loading.

$$W_c = \pi * \sigma_o^2 * \left(\frac{1}{\frac{G^*}{\sin(\delta)}} \right)$$

§ Therefore, to minimise W_c , the parameter $[G/\sin(\delta)]$ must be **maximized**.

§ Hence, we want hard but elastic binder

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Fatigue Cracking

§ Distress in Wheelpath

§ Progressive Damage

- longitudinal cracking
- alligator cracking
- potholes

§ Affected by

- asphalt binder
- aggregates
- pavement structure



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§ For strain controlled loading, the dissipated energy can be expressed mathematically as:

$$W_c = \pi * \epsilon_o^2 * (G^* \sin(\delta))$$

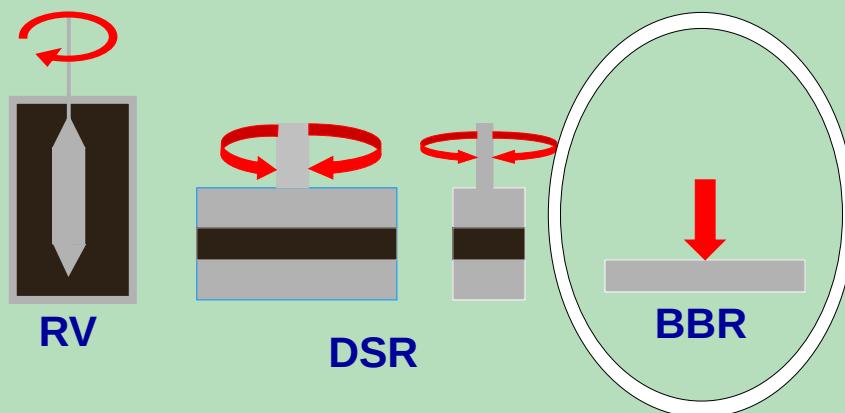
§ To minimise W_c , then the parameter $[G^* \sin(\delta)]$ must be **minimised**.

§ Either reducing G^* or reducing δ or both. This means we need soft but elastic binder.

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Thermal Cracking



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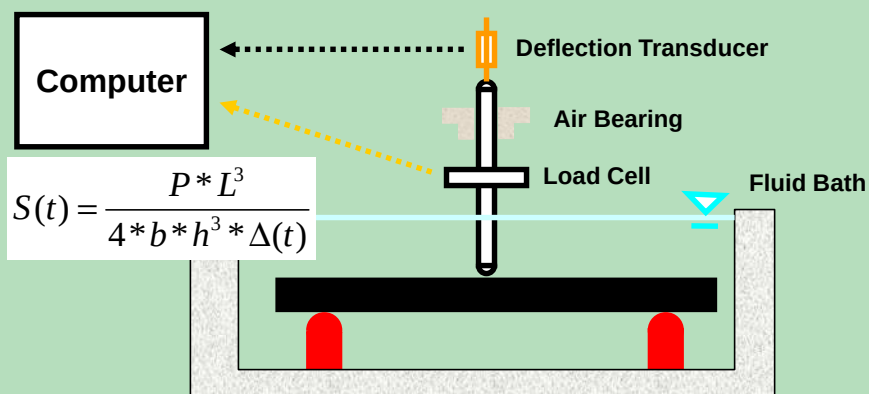
Bending Beam Rheometer BBR

- § BBR measures the stiffness of bitumens at very low temperatures.
- § The test uses engineering beam theory to measure the stiffness of a small bitumen beam sample under a creep load.
- § A creep load is used to simulate the stresses that gradually build up in a pavement when temperature drops.
- § Creep Stiffness and m-value are the two parameters evaluated with the BBR.

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Bending Beam Rheometer



$S(t)$ = stiffness (or Young's modulus) units kPa
 $D(t)$ = creep compliance = $1/S(t)$ units kPa^{-1}

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§ Creep Stiffness: Is a measure of the thermal stresses developed in the HMA as a result of thermal contraction. As $S(t)$ increases, the thermal stresses developed in the pavement due to thermal shrinking also increase, and thermal cracks becomes more likely.

§ $S(t) \leq 300 \text{ MPa}$

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§ m-value: indicates the rate of change of the creep stiffness w.r.t. loading time. It expresses the rate of stress relaxation.

§ Therefore, as m-value decreases the rate of stress relaxation decreases and the thermal cracks become more likely.

§ m-value ≥ 0.3

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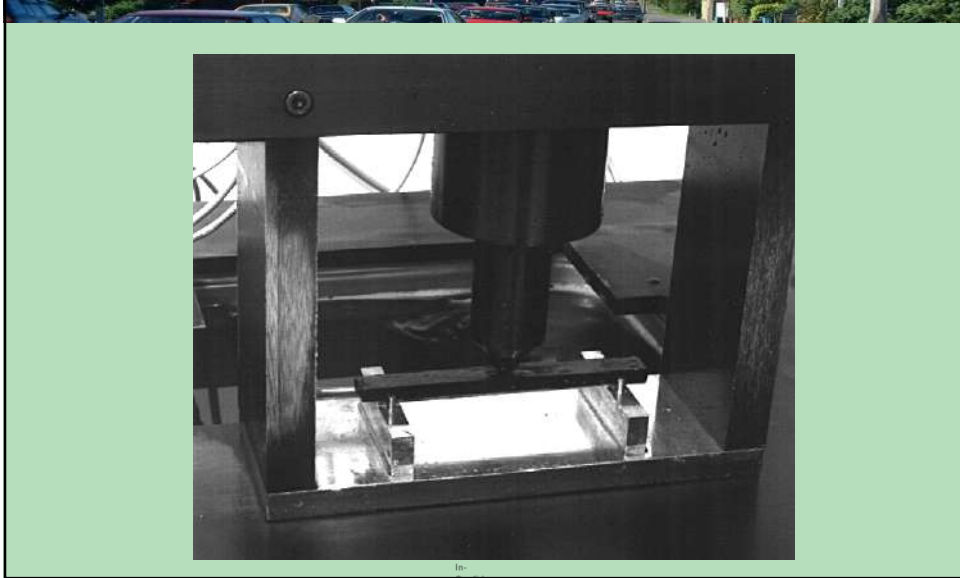
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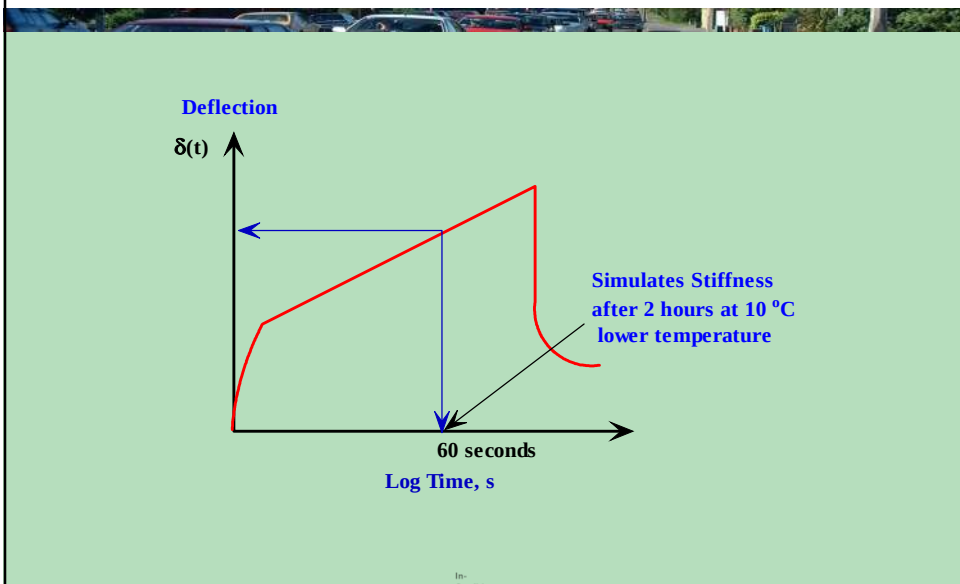


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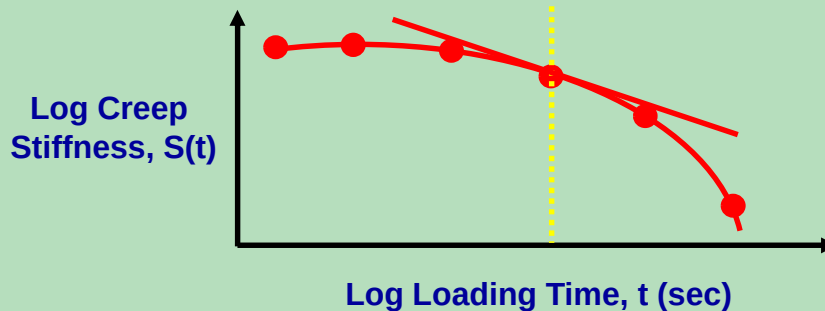


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Bending Beam Rheometer

§ Evaluates low temperature stiffness properties

- Creep stiffness
- Slope of response (called m-value)

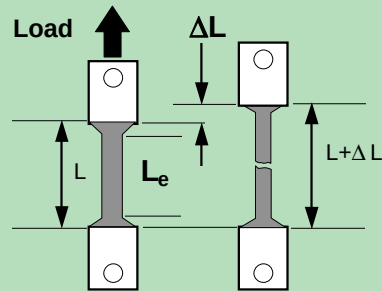


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Direct Tension Tester

- § Creep stiffness of binder as measured by the BBR is not adequate to completely characterise the low temperature behaviour.
- § Polymer modified binders may have $S(t)$ values greater than 300 MPa but do not crack because they can stretch further before breaking.
- § Carry out DTT only if $S(t)$ ranges between 300 to 600 MPa.

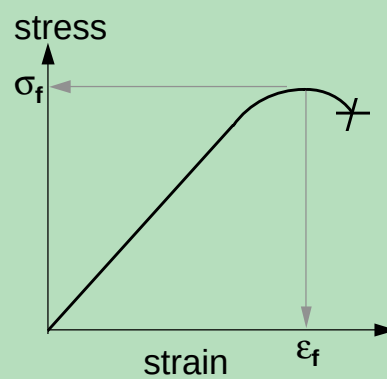
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$$\text{failure strain } (\epsilon_f) = \frac{\text{change in length } (\Delta L)}{\text{effective length } (L_e)}$$

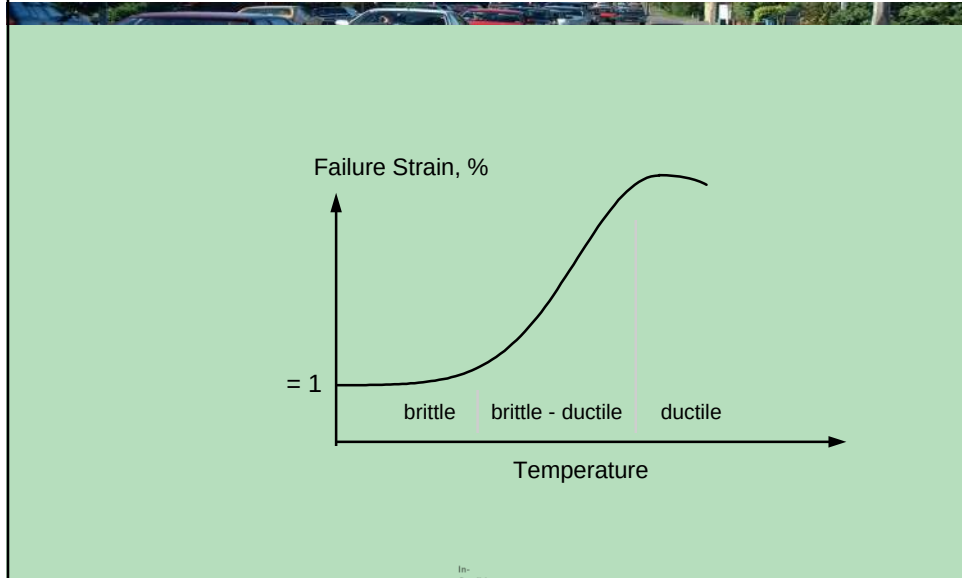
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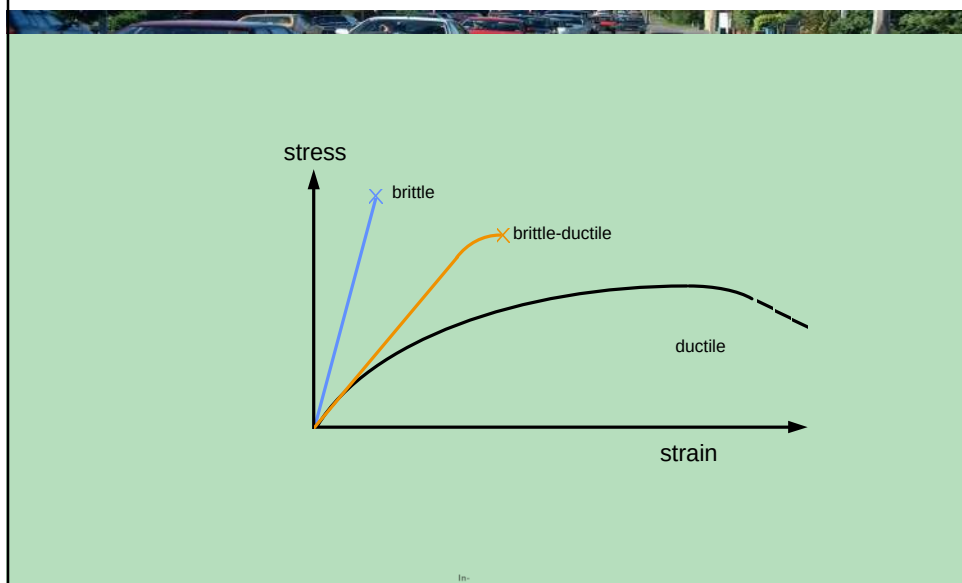


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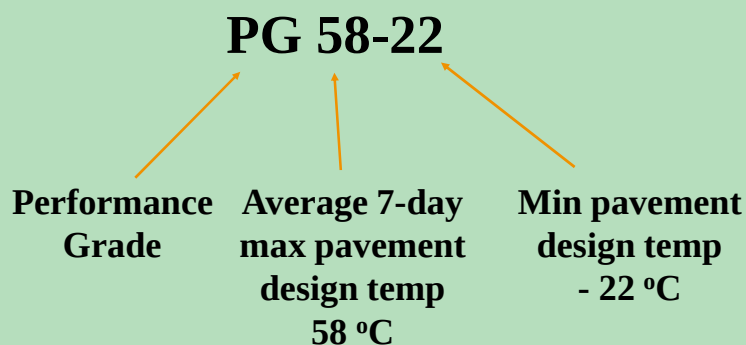
Superpave Specifications

- § Superpave specifications are performance based specifications. The measured test parameters are directly related to field performance.
- § By applying the superpave criteria, this will improve the performance of the mix and will limit the contribution of the bitumen binder to rutting, fatigue and thermal cracks.
- § Binders are selected and graded based on the actual pavement temperatures. Maximum 7 day pavement temperature and minimum pavement temperature.

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Superpave Performance Grades

§ Grading System Based on Climate



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Major differences between Superpave and Old Specs

§ The physical properties of the bitumen binder remain constant for all of the performance grades (PG), however, the temperatures at which these properties must be achieved, vary depending on the climate in which the binder is expected to serve.

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SUPERPAVE SPECIFICATIONS

Requirements - Constants

Temperature for tests

- * DSR - TP5
- * $G^*/\sin\delta > 1.00 \text{ kPa}$
- * $G^*/\sin\delta > 2.20 \text{ kPa}$
- * Temp. test @ 10 rad/sec.

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Max 7 Day Temp

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Min Pav Temp

Temp. test = Temp. of design

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High Temperature (Rutting) Parameter

§ The permanent deformation is caused by the dissipated energy (work done) with every cycle of traffic loading.

§ Therefore, to minimise permanent deformation, the dissipated energy in the pavement must be minimised.

$$W_c = \pi * \sigma_o^2 * \left(\frac{1}{\frac{G^*}{\sin(\delta)}} \right)$$

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§ To minimise the dissipated Energy we have to maximise this parameter ($G^*/\sin(\delta)$).

§ To maximise $G^*/\sin(\delta)$ either

- Increase G^* which means use stiff binder
- Decrease phase angle (δ) which means use Elastic Binder.

In-

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Permanent Deformation

Avg 7-day Max, °C	PG 46	PG 52	PG 58	PG 64	PG 70	PG 76	PG 82
1-day Min, °C	-34	-28	-22	-16	-10	-4	2
ORIGINAL							
(Flash Point) FP							
(Rotational Viscosity) RV							
(Dynamic Shear Rheometer) DSR $G^*/\sin \delta$							
	52	58	64	70	76	82	
(ROLLING THIN FILM OVEN) RTFO Mass Loss $\leq 1.00\%$							
(Dynamic Shear Rheometer) DSR $G^*/\sin \delta$							
	52	58	64	70	76	82	
(PRESSURE AGING VESSEL) PAV							
	90	90	100	100	100 (110)	100 (110)	110 (110)
(Dynamic Shear Rheometer) DSR $G^*/\sin \delta$							
	4	25	22	19	14	11	8
(Bending Beam Rheometer) BBR "S" Stiffness & "m" value							
	-24	-30	-34	-38	-42	-46	-50
(Bending Beam Rheometer) BBR Physical Hardening							
(Direct Tension) DT							
	-24	-30	-34	-38	-42	-46	-50

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Rutting Requirements

Viscosity, ASTM D 4402: ^b	
Maximum, 3 Pa-s (3000 cP),	
Test Temp, C	
Dynamic Shear, TP5: ^c	
$G^*/\sin \delta$, Minimum, 1.00 kPa	←
Test Temperature @ 10 rad/s, C	
Rolling Thin Film Oven (T240)	
Mass Loss, Maximum, %	
Dynamic Shear, TP5:	
$G^*/\sin \delta$, Minimum, 2.20 kPa	←
Test Temp @ 10 rad/sec, C	

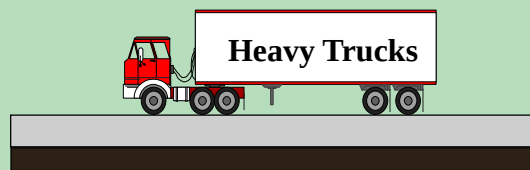
Spec Requirements to Address Rutting

98

Permanent Deformation Requirements

§ Addressed by high temperature stiffness

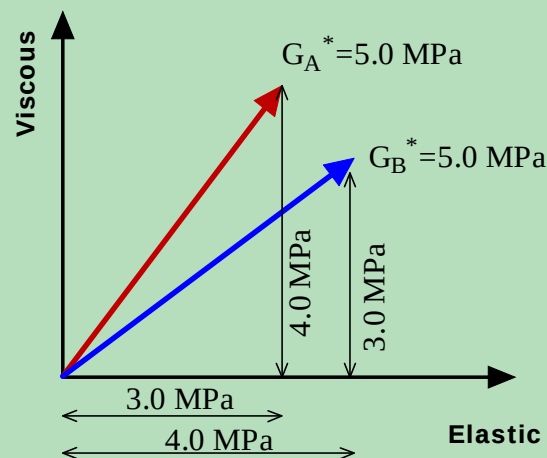
- $G^*/\sin \delta$ on unaged binder ≥ 1.00 kPa
- $G^*/\sin \delta$ on RTFO aged binder ≥ 2.20 kPa



> *Early part of pavement service life*

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Example



100

Fatigue Cracking

§ Distress in Wheelpath

§ Progressive Damage

- longitudinal cracking
- alligator cracking
- potholes

§ Affected by

- asphalt binder
- aggregates
- pavement structure



101

Intermediate Temp. parameter (Fatigue)

§ Fatigue cracking is typically considered a stress controlled phenomenon for thick pavement structures and strain controlled phenomenon for thin pavement structures.

§ Fatigue cracking is known to be more prevalent in thin pavements

§ Fatigue cracking always happens at later stage of the pavement service, therefore, binder must subject to long term ageing.

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In

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52

Fatigue Requirements

PAV Aging Temp, C

Dynamic Shear, TP5:

G* $\sin \delta$, Maximum, 5000 kPa

Test Temp @ 10 rad/sec, C

Physical Hardening^e

Creep Stiffness, TP1:^f

S, Maximum, 300 MPa

m-value, Minimum, 0.300

Test Temp, @60 sec, C

Direct Tension, TP3:^f

Failure Strain, Minimum, 1.0%

Test Temp @ 1.0 mm/min, C

Specification requirement
to address fatigue cracking

105

Low Temperature cracking requirements

§ When pavement temperatures drop down, pavement structure shrinks and tensile stress will develop in the pavement. The higher the creep stiffness, the higher the stress that will develop in the pavement and therefore the more likely it will crack.

§ Superpave limit the maximum value of creep stiffness to 300 MPa and the minimum value of m-value to 0.3

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PG 82

The image shows a close-up of a Dynamic Shear Rheometer (DSR) control panel. A magnifying glass is positioned over a yellow circular area that contains a red double-headed vertical arrow and the text "PAV Aged". The control panel features several digital displays and buttons. At the top, there are displays for "46", "62", and "18". Below these are three buttons labeled "90", "90", and "100". A large display shows "(PRESSURE AGING VESSEL)". Below this, a row of buttons is labeled "(Dynamic Shear Rheometer)". Further down, a row of buttons is labeled "(Bending Beam Rheometer)". At the bottom, a row of buttons is labeled "Beam Rheometer". The panel also includes a large "ON" button and a "STOP" button.

PG 82

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Low temperature cracking requirements

PAV Aging Temp, C
Dynamic Shear, TP5:
G* sin δ , Maximum, 5000 kPa
Test Temp @ 10 rad/sec, C
Physical Hardening ^e
Creep Stiffness, TP1: ^f
S, Maximum, 300 MPa
m-value, Minimum, 0.300
Test Temp, @60 sec, C
Direct Tension, TP3: ^f
Failure Strain, Minimum, 1.0%
Test Temp @ 1.0 mm/min, C

Specification requirements to address low temperature cracking

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Miscellaneous Spec Requirements

	PG 46	PG 52	PG 58	PG 64	PG 70	PG 76	PG 82
Flash Point	-4	-40	-40	-10	-28	-34	-40
Rotational Viscosity	22	28	34	40	46	52	58
Dynamic Shear Rheometer	52	58	64	70	76	82	88
(ROLLING THIN FILM OVEN)	90	100	110	120	130	140	150
Dynamic Shear Rheometer	52	58	64	70	76	82	88
Pressure Aging Vessel	90	100	110	120	130	140	150
20 Hours, 2.07 MPa	90	100	110	120	130	140	150
Dynamic Shear Rheometer	52	58	64	70	76	82	88
DSR G* sin δ	10	7	4	25	22	19	16
Bending Beam Rheometer	24	30	36	42	48	54	60
BBR "S" Stiffness & "m" value	24	30	36	42	48	54	60
Bending Beam Rheometer	24	30	36	42	48	54	60
Physical Hardening	24	30	36	42	48	54	60
Direct Tension	24	30	36	42	48	54	60
Report Value	24	30	36	42	48	54	60
≥ 1.00 %	24	30	36	42	48	54	60

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§ SuperPave Fundamentals

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Superpave Asphalt Binders

§ Grading System and Selection Based Primarily on Climate

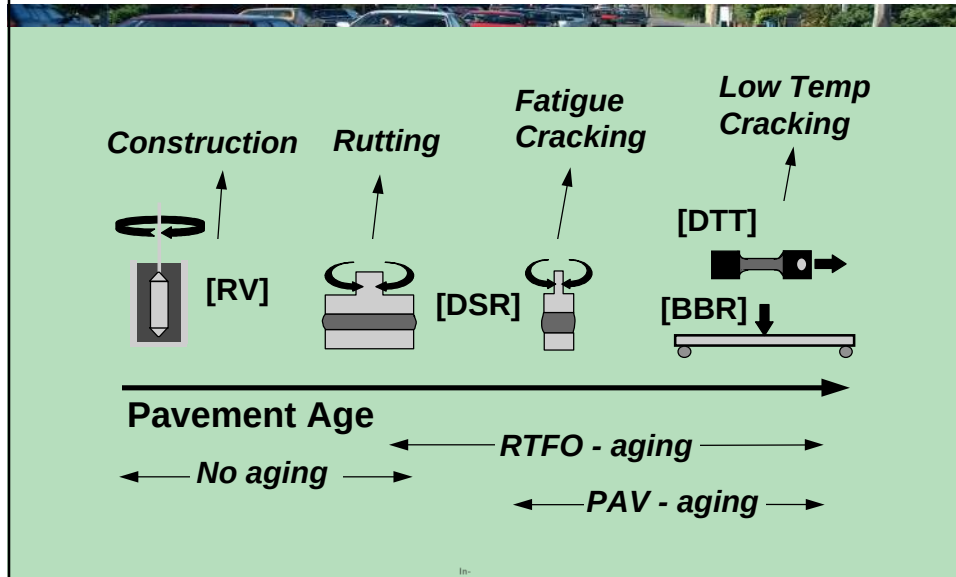
PG 58-22

Performance
Grade

Average 7-day
max pavement
design temp

Min pavement
design temp

112



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Developed from Air Temperatures

§ Weather Database of data collected over 20 years.

§ Annual air temperatures

- hottest seven-day temp (avg and std dev)
- coldest temp (avg and std dev)

§ Calculated pavement temperatures used in PG selection

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Air-Pavement Temperatures Relationship

$$T_{20mm} = (T_{air} - 0.00618 * Lat^2 + 0.2289 * Lat + 42.2) * 0.9545 - 17.78$$

T_{20mm} = Pavement Temperature at a depth of 20 mm in °C

T_{air} = Max. average high air temperature during the hottest seven-day period in °C

Lat = Project Latitude in degrees

$$T_{min} = 0.859 * T_{air} + 1.7$$

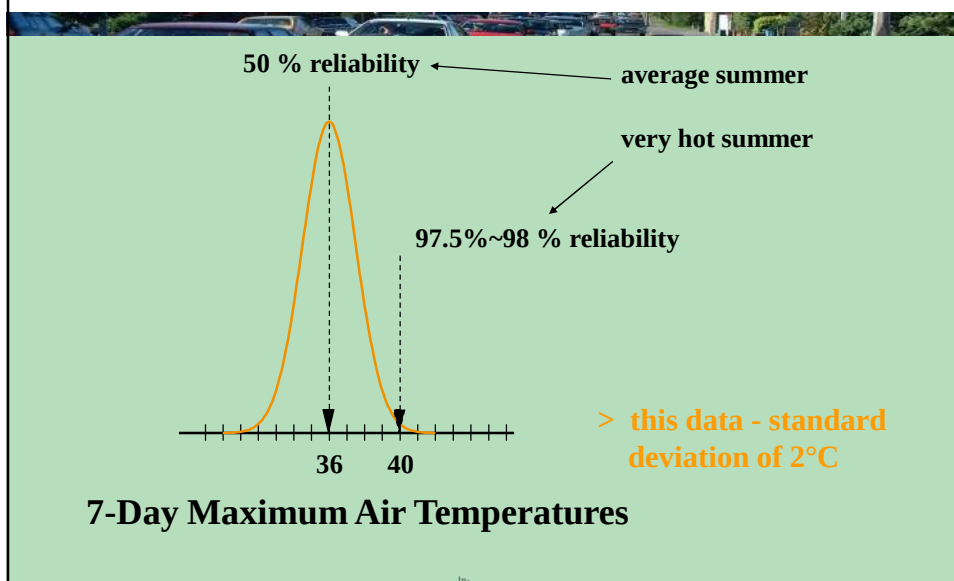
T_{min} = Minimum pavement temperature in °C

T_{air} = Min air temperature in average year in °C

In-

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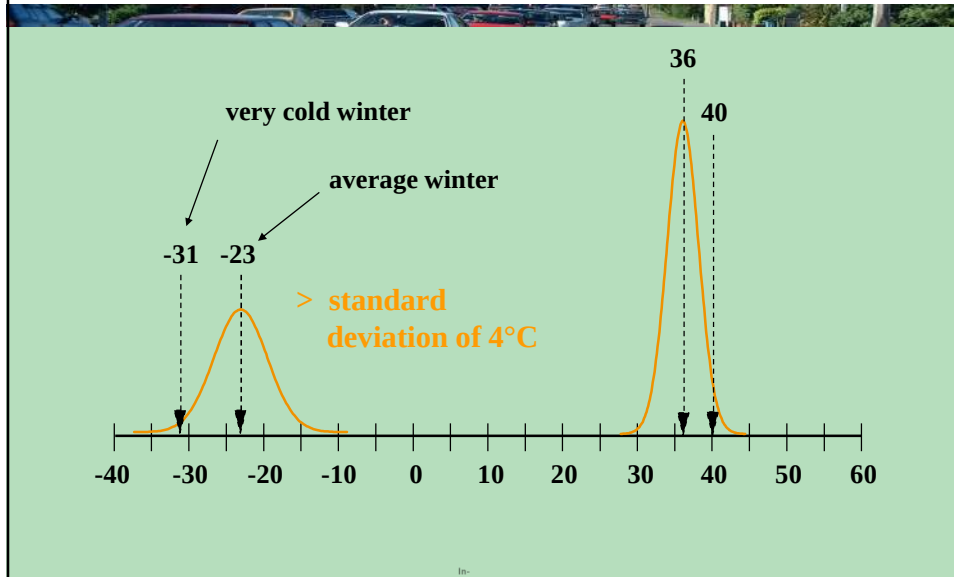
Observed Air Temperatures



In-

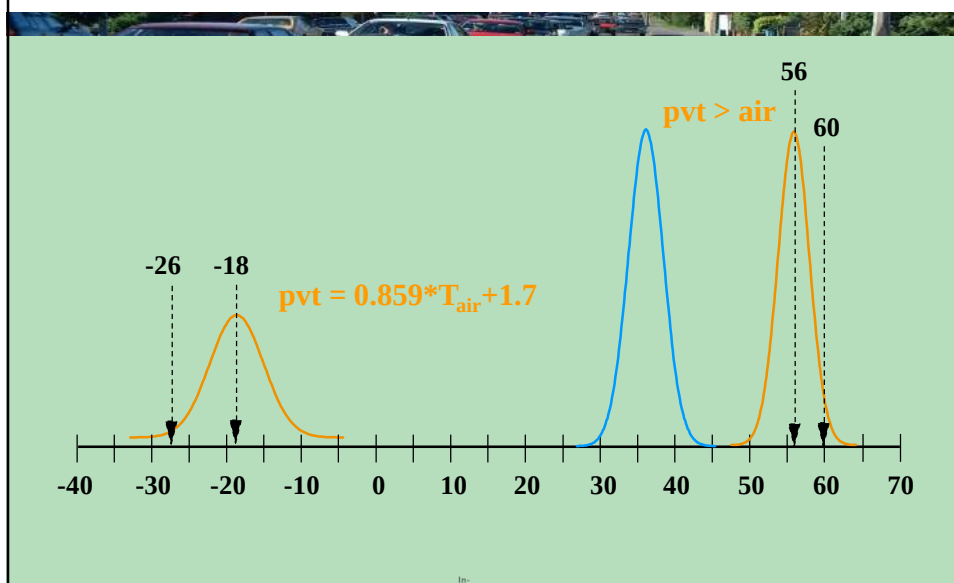
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Observed Air Temperatures



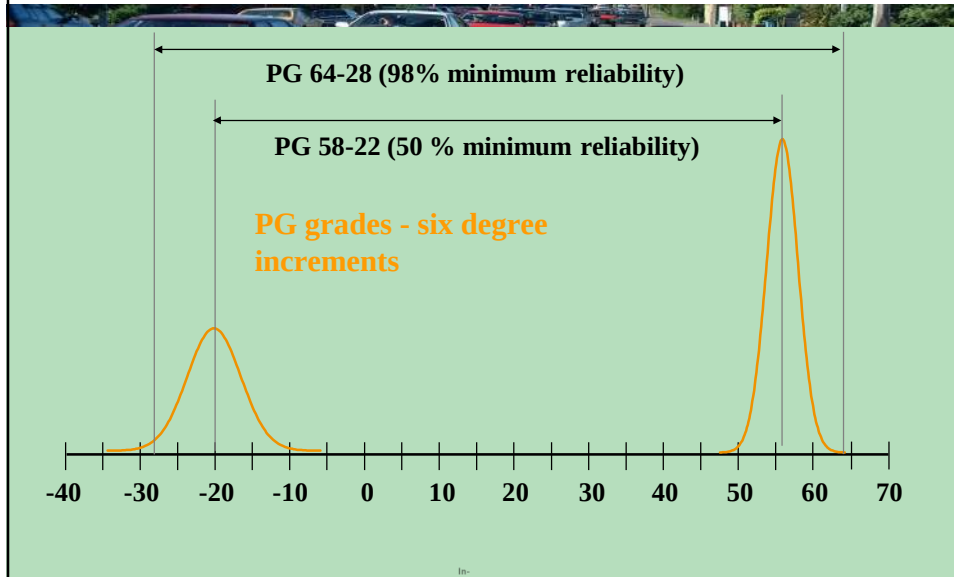
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Calculated Pavement Temperatures



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PG Binder Grades



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Superpave Binder Grades

High Temperature Grade	Low Temperature Grade
PG46-	34,40,46
PG52-	10,16,22,28,34,40,46
PG58-	16,22,28,34,40
PG64-	10,16,22,28,34,40
PG70-	10,16,22,28,34,40
PG76-	10,16,22,28,34
PG82-	10,16,22,28,34

NZ PG grades

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Summary of Superpave Tests for a PG 64-22

Test	Construction	Permanent Deformation (Rutting)		Fatigue Cracking	Low-Temperature Cracking	
		DSR		DSR	BBR	
Aging Condition	None	None	RTFO	RTFO + PAV	RTFO + PAV	
Test Temp.	135°C	Max. Pavement Temp. (64°C)	Max. Pavement Temp. (64°C)	0.5 (Max. Pavement Temp. + Min. Pavement Temp.) + 4 (25°C)	Min. Pavement Temp. + 10°C (-12°C)	
PG 64-22						
Parameter	Visc. ≤ 3 Pa-sec	$ G^* /\sin \delta \geq 1.0$ kPa	$ G^* /\sin \delta \geq 2.2$ kPa	$ G^* \times \sin \delta \leq 5000$ kPa	$S(t = 60 \text{ sec}) \leq 300$ MPa	$m(t = 60 \text{ sec}) \geq 0.3$

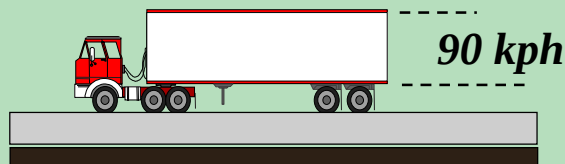
In-

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Superpave Asphalt Binders

§ Binder Selection Also Based on

- traffic speed
- amount of traffic



In-

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Effect of Loading Rate on Binder Selection




§ **Example**

- for toll road PG 64-22
- for toll booth PG 70-22
- for weigh stations PG 76-22

90 kph

Slow

Stopping

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Effect of Traffic Amount on Binder Selection




§ **10 - 30 x 10⁶ ESAL**

- Consider increasing - - one high temp grade

§ **30 x 10⁶ + ESAL**

- Recommend increasing - - two high temp grade

> Equivalent Single Axle Loads

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DESIGN ESA (million)	Adjustment to Binder PG Grade		
	Traffic Load Rate		
	Standing	Slow	Standard
< 0.3	-	-	-
0.3 to < 3	2	1	-
3 to < 10	2	1	-
10 to < 30	2	1	-
> 30	2	1	1

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Current AASHTO Binder Standards

Key AASHTO Standards for Superpave Binders

- **AASHTO M 320:** Standard Specification for Performance-Graded Asphalt Binder (the primary spec).
- **AASHTO R 29:** Standard Practice for Grading or Verifying the Performance Grade of an Asphalt Binder.
- **AASHTO M 323:** Standard Specification for Superpave Volumetric Mix Design (covers binder selection within the mix).
- **AASHTO T 315 (DSR):** Determining Rheological Properties of Asphalt Binder Using Dynamic Shear Rheometer.
- **AASHTO T 313 (BBR):** Determining Flexural Creep Stiffness of Asphalt Binder Using Bending Beam Rheometer.
- **AASHTO T 316 (Rotational Viscometer):** Viscosity Determination of Asphalt Binder Using Rotational Viscometer.
- **AASHTO T 240 (RTFOT):** Effect of Heat and Air on a Moving Film of Asphalt Binder (Short-Term Aging).

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